

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Unit 4 assessment

Unit summary

The threats that reach ordinary users are malware (virus, ransomware, spyware), phishing, and social engineering, and most are cast wide rather than targeted; the defence is a set of habits - update, verify before tapping, install from official sources, back up away from the device - protecting the CIA triad of confidentiality, integrity and availability. A digital footprint is the trail you leave online, active (what you post) and passive (what is collected), and teachers are searched more than most, so it is worth auditing: what is already there, what keeps growing, what you put out deliberately. A password is a secret kept by length, variety, no reuse and no sharing, with a password manager carrying the load; authentication rests on something you know, have or are, and multi-factor authentication combines two kinds so a phished password alone opens nothing. The cloud is other people's computers - on-demand, networked, pooled, elastic, metered - convenient and durable at the price of trusting the provider. Personal data belongs to the person it describes; a school is a caretaker of a community's richest data, most of it about children, and every collection must survive four questions: why, where, who, how long.

Summative assessment

Multiple-choice questions (MCQs)

1. Ransomware is best described as malware that: (*Remember*) a) spreads through email attachments only b) locks files and demands money to unlock them c) shows unwanted advertisements d) speeds up the computer's fan
2. A message that looks like it is from your bank and leads to a fake page that captures your password is: (*Understand*) a) a virus b) phishing c) spyware d)

a software update

3. The strongest password of these four is: (*Apply*) a) school123 b) Shakeel2010 c) Pk!9#mQ2vLx d) password2026
4. The "C" in the CIA triad stands for: (*Remember*) a) complexity b) confidentiality c) connectivity d) cloud computing
5. The passive part of a digital footprint is: (*Understand*) a) the posts you write b) the photos you upload c) the data collected about you as you use the internet d) the reviews you publish
6. A future employer searches your name online. Which trace is passive? (*Apply*) a) your blog post about teaching b) your location data collected by an app c) your comment on a school's page d) your profile photograph
7. Multi-factor authentication defeats a phished password because: (*Understand*) a) the password becomes longer automatically b) the attacker still needs the second factor, which is in your pocket c) the system locks the account forever d) the attacker's email is blocked
8. According to the NIST definition, cloud computing is: (*Remember*) a) weather forecasting with computers b) on-demand network access to a shared pool of configurable computing resources c) a local backup drive d) a type of antivirus
9. A vendor asks for pupils' home addresses for an app that works without them. Under collect-the-minimum, the school should: (*Apply*) a) give everything to be safe b) give only the data the app actually needs c) refuse the app entirely without asking why d) ask the pupils to decide
10. Which of these is sensitive personal data in a school? (*Remember*) a) the school's timetable b) a pupil's health notes c) the school's noticeboard d) the name of the textbook

Restricted-response questions (RRQs)

1. Describe the three families of cyber threats, giving one example of each. (*Understand*)

2. Explain the difference between phishing and social engineering in two sentences. *(Understand)*
3. A message says "verify your account in 2 hours or it closes". List three phishing tells and the action you would take instead of tapping. *(Apply)*
4. Explain why "my data is not worth stealing" is a false comfort for a teacher. *(Understand)*
5. Distinguish the active from the passive digital footprint with one example of each. *(Understand)*
6. Rank these three passwords from weakest to strongest and justify each verdict: school123, Shakeel2010, Pk!9#mQ2vLx. *(Analyze)*
7. Your email is phished but has MFA enabled. Describe what happens when the attacker tries to log in, and why. *(Apply)*
8. Explain the three kinds of authentication evidence, giving one example of each. *(Understand)*
9. Explain, in two sentences, why the cloud is described as "other people's computers" and what the trade is. *(Understand)*
10. A school keeps its only backup drive in the same cupboard as its computer. Analyse the risk using one threat from Topic 4.1. *(Analyze)*

Extended-response questions (ERQs)

1. Prepare a one-page "safe messages" guide for the school staff room: the three threat families in plain words, five phishing tells, and the school's rule for messages that ask for a password or money, with a marked example. *(Analyze)*
2. Write a one-page footprint audit for yourself as a future teacher: what a search of your name would show, which traces help and harm, and three actions - clean, curate, prevent. *(Evaluate)*
3. The school office shares one password among five staff and keeps its only backup in the same cupboard as the computer. Recommend a security upgrade covering passwords, a second factor, and backups, tying each to a named threat. *(Analyze)*
4. Write a one-page data-protection plan for your practicum school: five kinds of data it holds, where each lives, the riskiest practice, and three rules you would introduce first. *(Analyze)*

5. Using the whole unit, evaluate the statement: "Online safety is a set of habits, not a product you can buy." Draw on the threat families, the CIA triad, authentication and data protection as evidence. (*Evaluate*)

Answers and marking guidance

MCQ answer key

1. b - ransomware locks and demands payment; that is its defining move.
2. b - the fake trusted message leading to credential capture is phishing by definition.
3. c - twelve characters, mixed types, no words; the others are dictionary words, names with years, or the literal worst password.
4. b - confidentiality: access limited to those allowed.
5. c - the passive footprint is collected about you; the others are deliberate.
6. b - location data collected by an app is passive; all the others are published by you.
7. b - the second factor is something you have; the phished password alone is not enough.
8. b - that is the NIST definition's core.
9. b - collect the minimum: give only what the app actually needs.
10. b - health notes are sensitive; the others are not personal data at all.

RRQ model answers and mark schemes

1. Model: malware (ransomware); phishing (fake bank message); social engineering (pretexting call to a helpdesk). **Marks (3):** 1 per family described with an example.
2. Model: phishing uses a fake message or site to steal credentials directly; social engineering is any attack that manipulates the person - phishing is one form of it, and pretexting works by voice with no software at all. **Marks (2):** 1 for each distinction correctly drawn.
3. Model: urgency ("2 hours"); a link instead of a known channel; generic greeting or wrong sender address; the action: do not tap, contact the bank

through its official number or app. **Marks (3):** 2 for three tells; 1 for the safe action.

4. Model: a teacher's account is valuable for sending more phishing in their name; marks sheets and contact lists harm others, not just you; and teachers are searched and judged professionally. **Marks (2):** 1 per valid reason.
5. Model: active is what you deliberately publish (a post); passive is what is collected as you use the internet (location or search history). **Marks (2):** 1 per side with an example.
6. Model: school123 weakest (dictionary word plus digits, tried first); Shakeel2010 weak (name plus year, guessable from public information); Pk!9#mQ2vLx strongest (twelve characters, mixed types, no words). **Marks (3):** 1 per verdict with its reason.
7. Model: the password is accepted as the first factor, then the system requests the second - a code on your phone; the attacker cannot complete login without the device, and you receive a code you did not request, which is itself a warning. **Marks (3):** 1 for the sequence; 1 for the pocket explanation; 1 for the warning sign.
8. Model: something you know (password), something you have (phone or security key), something you are (fingerprint), each explained with its example. **Marks (3):** 1 per kind with an example.
9. Model: cloud services run on the provider's machines in data centres, not yours; the trade is convenience and durability for trust in the provider with your data. **Marks (2):** 1 for whose computers; 1 for the trade.
10. Model: ransomware encrypts the computer and the backup beside it - both are lost together; the offline or cloud copy the family needed is in the same locked box as the machine. **Marks (3):** 1 for naming the threat; 2 for the analysis of the shared fate.

ERQ rubrics

1. **Safe-messages guide (Analyze).** Criteria: *Three threat families explained plainly (0-3); Five tells, each explained (0-3); The rule is actionable and the example's tells marked correctly (0-3, Analyze); One page, staff-room language (0-1).* Total 10.
2. **Footprint audit (Evaluate).** Criteria: *Describes actual trace types (0-3); Help and harm judged with reasons tied to teaching (0-4, Analyze); Three*

concrete actions naming settings or habits (0-2); Honest and clear (0-1).
Total 10.

3. **Security upgrade (Analyze).** Criteria: *Password problem diagnosed with its real risks (0-3); MFA recommendation explains the mechanism (0-3, Analyze); Backup recommendation addresses the same-location failure (0-2); Each tied to a named threat; clear (0-2).* Total 10.
4. **Data-protection plan (Analyze).** Criteria: *Inventory concrete and complete (0-3); Riskiest practice identified with its harm path (0-3, Analyze); Three actionable rules, one tied to a threat (0-3); One page (0-1).* Total 10.
5. **Integrative evaluation (Evaluate).** Criteria: *Draws on at least three of: threat families, CIA triad, authentication, data protection (0-3); Each drawn part used accurately (0-3); The habits-not-products judgement defended as a synthesis (0-3, Analyze); Clarity (0-1).* Total 10.

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